



Acoustic and Linguistic Biomarkers for Cognitive Impairment Detection from Speech

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Abstract

This work describes a comprehensive approach for the automatic assessment of cognitive decline from spontaneous speech in the context of the PROCESS Challenge 2025. Based on our previous experience on the use of speech and text-derived biomarkers for disease detection, we evaluate here the use of knowledge-based acoustic and text-based feature sets, as well as LLM-based macro-descriptors, and multiple neural representations (e.g., Longformer, ECAPA-TDNN, and Trillsson embeddings). The combination of these feature sets with different classifiers resulted in a large pool of systems, from which, those providing the best balance between train, development, and individual class performance were selected for model ensembling. Our final best-performing systems correspond to combinations of models that are complementary to each other, relying on acoustic and textual information from the three clinical tasks provided in the challenge dataset.

Index Terms: Pathological speech, dementia detection, PROCESS challenge, speech embeddings, macro-descriptors

1. Introduction

Dementia, which affects fifty-five million people around the world, is one of the major causes of disability and dependency among older people [1]. It is marked by a progressive decline in cognitive functions beyond what is considered normal in biological aging [1]. Alzheimer’s disease (AD), a progressive neurodegenerative disorder, is the most common form of dementia. Although memory impairment is the most prominent symptom, speech and language impairments are also persistent and useful in detecting early onset of the disease [2, 3, 4]. Mild cognitive impairment (MCI) is a clinical condition that often leads to dementia. It is characterized by noticeable deficits in one or more cognitive domains that exceed typical age-related changes but do not meet the diagnostic criteria for dementia [5, 6].

While numerous studies have leveraged speech and language biomarkers for detecting dementia, AD, and MCI (e.g., [7, 8, 9, 10, 11, 12]), this field remains hindered by a lack of standardization and sufficiently large, representative benchmarks for systematic comparison of methodologies. Notable exceptions include the ADReSS Challenges, which focus on AD detection [13, 14, 15], and the Taukadi Challenge, which incorporates data in English and Mandarin to address MCI detection [16]. Recently, the PROCESS Challenge introduced a dataset designed for three-class classification, distinguishing healthy controls, individuals with MCI, and patients with dementia, a problem setting that is less commonly studied [17].

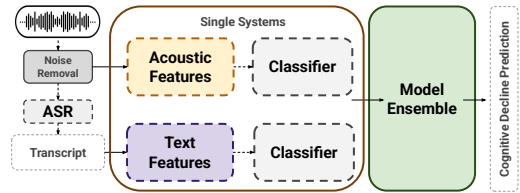


Figure 1: Overall schema of our proposed method.

In this paper, we describe our participation in the PROCESS Challenge 2025 where, building on our previous work, we proposed and experimentally evaluated diverse approaches for distinguishing healthy controls, individuals affected by MCI, and dementia patients. Namely, we explored complementary data representations targeted at capturing distinct manifestations (i.e., potential symptoms) of early cognitive impairment often reflected in speech and language production. Such representations include knowledge-based (KB) acoustic and linguistic features, text- and speech-based neural embeddings, and macro-descriptors derived from LLMs, which were combined with multiple classification strategies. The resulting top-performing systems were fused using logistic regression to achieve two final systems. This fusion strategy allowed us to leverage complementary information and substantially improve the final classification performance. The diagram in Figure 1 summarizes our approach. Additional details on feature extraction and results are available at github.com/mccatarinatb/IS-25-supply-material.

2. Corpus

The PROCESS Challenge dataset [17] was designed to promote research on early detection of dementia via speech analysis.

Dataset Overview. The corpus includes three diagnostic classes: (i) the **Healthy Control (HC)** group includes volunteers without cognitive impairment, and individuals possibly experiencing memory issues unrelated to neurodegenerative conditions; (ii) the **MCI** group; and (iii) individuals diagnosed with early-stage **Dementia**, exhibiting varying degrees of cognitive decline. Detailed demographic and disease severity statistics for the training and development sets are reported in Table 1.

Cognitive Assessment Scores. The dataset also offers the Mini-Mental State Examination (MMSE) score for a subset of participants. The MMSE, commonly used to assess cognitive impairment, ranges from 0 to 30 and attains lower scores in individuals who completed comparatively fewer years of education and/or have diagnosable diseases [18].

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Table 1: Demographic and severity statistics of the official training and development sets of the PROCESS Challenge.

Group	Gender	#Subjects		Age		MMSE Score [†]	
		train	dev	train	dev	train	dev
HC [†]	Male	23	12	64.8±13.3	63.6±13.1	29.0±1.0	29.2±1.2
	Female	37	9	62.9±12.1	61.0±14.3	28.9±0.7	29.5±0.5
MCI	Male	22	7	69.1±7.9	68.1±10.8	27.1±1.8	23.7±3.4
	Female	22	8	69.8±10.3	61.6±13.0	26.5±2.6	27.0±1.6
Dementia	Male	8	3	75.8±8.0	67.0±3.7	26.3±2.1	27.7±0.9
	Female	4	1	69.0±6.3	60.0±0.0	20.0±0.0	—

[†] Gender was not reported for 1 subject but included in our experiments.

[‡] The MMSE score was only provided for a subset of 69 subjects.

Cognitive Assessment Tasks. The corpus includes audio recordings from three types of elicitation tasks, resulting in three audio files per speaker:

- **Cookie Theft Description (CTD).** Subjects describe the well-known “Cookie Theft” picture [19], to assess cognitive functions such as language comprehension and memory.
- **Phonemic Fluency (PFT).** Subjects utter as many words starting with the letter “P” as possible within a minute, to evaluate executive function and phonological processing.
- **Semantic Fluency (SFT).** Subjects are asked to list as many animals as possible in one minute, to assess verbal fluency, lexical access, and semantic memory.

3. Methodology

Figure 1 shows an overview of the proposed method. The speech signals are pre-processed to remove noise prior to automatic transcription. Features are then extracted from both the cleaned speech signals and their corresponding transcripts, serving as inputs for classifier training. Finally, the best-performing individual systems are combined through model ensembling to generate the final predictions. This fusion of complementary information helps tackle the challenges faced in accurately classifying dementia cases in this dataset.

3.1. Pre-processing

A preliminary inspection of the recordings identified two undesired factors: the presence of loud noises (signaling tones) and the interventions of interviewers. The former was dealt with through signal processing approaches while, for the latter, we experimented with speaker diarization systems for the automatic removal of those interventions. However, early experiments showed that state-of-the-art diarization systems had very poor performance in this setting, often removing large segments corresponding to the subject. For this reason, and after a preliminary evaluation of manual removal of interviewers’ interventions that showed little difference in performance, we opted to skip this step.

Noise Removal. Several recordings in the challenge dataset contain signaling tones from the recording protocol that marked the beginning and the end of the tasks, in addition to various other loud noises. To prevent such noises from interfering with the detection of cognitive impairment, an energy-based denoising step was introduced to replace those sounds with low-energy random Gaussian noise in segments without speech.

Automatic Speech Recognition. We compared two versions of OpenAI’s Whisper [20], *whisper-medium.en* and *whisper-large-v2*, as well as CrisperWhisper [21], a variant of Whisper designed to capture every spoken element, including disfluencies. This characteristic makes CrisperWhisper a promising tool in the study of neurodegenerative disorders, where disfluencies hold diagnostic information.

Table 2: ASR performance (WER [%]) per speech task, on train+dev sets, for which manual transcripts were available

	All	CTD	PFT	SFT
<i>whisper-large-v2</i>	35.9	31.9	49.7	41.3
<i>whisper-medium.en</i>	26.3	19.3	43.7	39.6
<i>CrisperWhisper</i>	23.2	14.1	54.7	34.8

Table 2 presents the word error rate (WER) performance of each model after a post-processing step, where disfluencies were standardized with a uniform placeholder to reduce the impact of non-standard spellings. The table highlights that fluency tasks are more challenging than CTD, likely because the ASR systems were trained on continuous speech following grammatical rules, while fluency tasks correspond to word listings with very different structure. Despite some hallucinations in fluency tasks, CrisperWhisper achieved lower WER than Whisper for CTD and SFT, and was thus used for their transcripts, while *whisper-medium.en* was used for PFT samples.

3.2. Feature Extraction

To model manifestations of cognitive impairment, we used feature representations derived from both the acoustic and linguistic components of speech, including neural and KB features.

Acoustic features. In terms of KB features, we compared four different sets: eGeMAPS [22] and ComParE [23], both extracted with OpenSMILE [24]; a set of 33 acoustic features extracted with *Praat* [25], henceforth referred to as *Praat*; and a subset of these, containing 11 pause-related features, referred to as *Pauses*. The *Praat* features strongly overlap with those described in [26]. In terms of neural speech representations, we used ECAPA-TDNN speaker embeddings (ECAPA) [27] and TRILLsson paralinguistic embeddings [28].

Text features. We compared two different KB feature sets: one set of 11 linguistic features (*Ling*) proposed by [26], and one set of 8 linguistic features designed for fluency tasks (*Fluency*). Additionally, we explored the macro-descriptors (*Macro*) proposed in [29], estimated using Llama-3.1-70B-Instruct [30]¹. In terms of neural representations, we explored embeddings from three pre-trained Language Models (PLMs), namely BERT [31], RoBERTa [32], and Longformer [33].

3.3. Classification

For classification, we use Support Vector Machines (SVMs), Decision Trees (DT), Random Forest (RF), and Multi-layer Perceptron (MLP). Besides, we explore Fuzzy Fingerprints (FFP), an interpretable classification technique that has proved to be useful when there is high class imbalance in text datasets [34].

To address data scarcity, the training set was divided into five folds, ensuring a balanced distribution of classes and gender. Classifiers were trained using five-fold cross-validation, and predictions for the development set were obtained by averaging the outputs of the five classifiers. These averaged predictions were then used to compute the final performance metrics. All systems were evaluated in terms of unweighted average F1-score (UAF1), following the challenge guidelines. The F1-score for each class is also reported. As the official test set was unavailable, performance is reported only for the training set (using cross-validation) and the held-out development set.

While most systems were trained for each of the three tasks separately, preliminary experiments showed that acoustic neural embedding-based systems benefited from a combination of the three speech tasks to obtain speaker-level predictions.

¹Large-language models that are only available through online APIs were not used due to restrictions imposed by the challenge’s organizers.

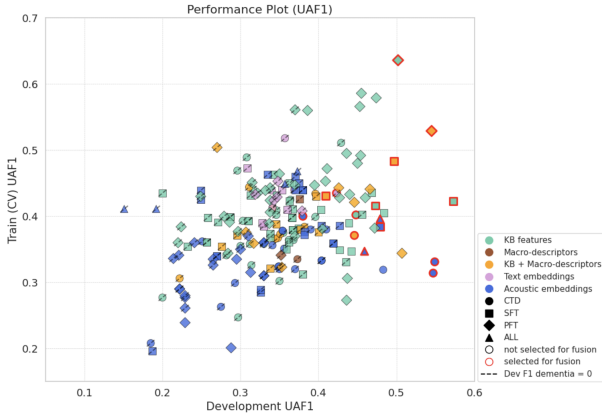


Figure 2: Single systems performance, in terms of UAF1.

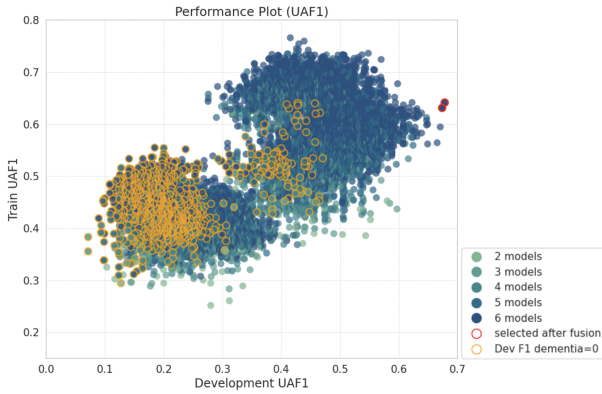


Figure 3: Model ensemble performance, in terms of UAF1.

3.4. Model Ensemble

To exploit the potential complementary information embedded in individual systems (i.e., one or more feature sets followed by a classifier), we adopted a late fusion strategy. After evaluating every single system, the top-performing ones were selected, with a focus on those providing complementary information while exhibiting a balanced performance across the cross-validation and development sets, as well achieving non-zero F1-scores for the dementia class whenever possible. Then, the prediction scores provided by each system for each class – either as probabilities, softmax scores or, in the case of SVMs, distances to the class-separating hyperplane – were aggregated using a logistic regression model. Specifically, for each target class, system-dependent scaling factors and offsets were estimated and calibrated to approximate log-likelihood ratios using the training scores obtained in cross-validation [35]. Then, the fusion models were applied to the development set and the class with the maximum log-likelihood ratio score was the predicted one. In these experiments, we considered all possible ensemble combinations of up to six single systems at a time.

4. Results & Discussion

Single-system experiments. The results for the individual systems are reported in Figure 2, where the best-performing models can be found in the top right corner. This figure shows how systems trained on the PFT task generally achieve the highest performance. Moreover, KB features and their combination with macro-descriptors consistently yield superior results, when compared to other data representations. The figure also

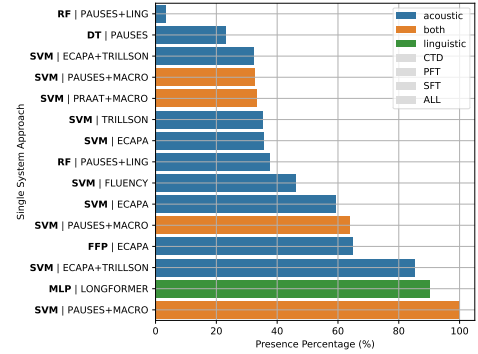


Figure 4: Frequency of single classifiers appearing in the top best-performing ensemble model combinations, categorized by the type of features and tasks considered during their training.

shows that several systems, despite showing reasonable UAF1, attain an F1-score of 0 in the dementia class, showcasing the challenging nature of this dataset. Possible reasons for this include the age discrepancies observed for the dementia subjects in the training and development partitions, and the significant class imbalance in the corpus.

Of the 205 individual systems evaluated, 15 were identified as promising candidates for fusion and subsequently considered in the model ensemble stage. These systems, marked with a red edge in Figure 2, include both top- and medium-performing models. This diverse selection of approaches aimed to incorporate complementary information, potentially enhancing the overall effectiveness of the ensemble.

Model ensemble experiments. The 15 systems selected were combined using the ensemble strategy described in Section 3.4. By defining combinations of 2 up to 6 systems at a time, we explored around 10k system combinations in our experiments.

The performance of these model ensembles is presented in Figure 3. In this figure, it is possible to observe that a large number of systems are capable of generalizing from training to development. In general, ensembles including a larger number of single systems seem to correlate with a higher performance on both partitions, indicating that the proposed approaches provide complementary information. Furthermore, the fusion of models through logistic regression provides a substantial performance improvement when compared to the individual models shown in Figure 2. Still, we observe that several models, highlighted in yellow, exhibit an F1-score of 0 for the dementia class.

Fine-grained analysis. For a more refined analysis, we selected 421 ensembles that achieved a UAF1 score above 55% in the training and development sets and a non-zero F1 score for the dementia class. Figure 4 illustrates the frequency with which each of the 15 individual systems appears in this set. One key observation is that the system that incorporates pause features (*Pauses*) and macro-descriptors (*Macro*) trained on the PFT task is consistently selected, with the same system trained on SFT samples being selected over 60% of the time. Additionally, the system based on Longformer embeddings for the CTD task and the system based on the combination of ECAPA and TRILLSSon embeddings for all tasks were selected in more than 80% of the best-performing model ensembles.

These observations confirm the relevance of both acoustic and linguistic modalities for this task, from which complementary data representations can be extracted to capture distinct manifestations of early cognitive impairment. Not only neural representations, but also interpretable high-level concepts such

Table 3: Systems selected for the best-performing model ensembles (Ensemble#1 and Ensemble#2).

System #	Data representation	Task	Classifier	Ensemble#1	Ensemble#2	Description
1	Longformer	CTD	MLP	–	✓	This system uses a fine-tuned version of the pre-trained Longformer [33], whose modified attention mechanisms on both local and global scales allow for tackling longer texts. Input transcriptions are fed to the Longformer, the $[CLS]$ token is used as pooling strategy, and a fully-connected linear layer is used for classification.
2	Fluency	SFT	SVM	✓	✓	First, transcripts are processed by an LLM (Llama-3.1-70B-Instruct), prompted to identify all “target words” corresponding to animals. Then, six fluency features are computed, including counts and repetitions of unique target words, as well as the mean and standard deviation of cosine similarities between Word2Vec embeddings of adjacent target words. These features are fed into an SVM with a linear kernel and $C=0.1$.
3	Pauses + Macro	SFT	SVM	✓	✓	The set of eleven pause related features and four macrodescriptors is fed to an SVM with a linear kernel and $C=0.1$, resulting in two systems, one for PFT and another for SFT.
4		PFT	SVM	✓	✓	
5	ECAPA	SFT	FFP	✓	–	ECAPA-TDNN embeddings are extracted from the SFT task, and Fuzzy Fingerprints used for classification.
6	ECAPA	All	SVM	✓	✓	This system uses a linear SVM with $C=0.0001$, trained over the concatenation of ECAPA-TDNN embeddings, for all three tasks. PCA dimensionality reduction is applied before classification.
7	ECAPA TRILLsson	+ All	SVM	✓	✓	Similar to the system above, but using both the PCA-reduced ECAPA-TDNN and TRILLsson embeddings in concatenation.

Table 4: F_1 -score (%) analysis for the best-performing model ensembles and the single systems comprising them.

System #	Cross-validation				Held-out Development			
	UAF ₁	F ₁ HC	F ₁ MCI	F ₁ Dem	UAF ₁	F ₁ HC	F ₁ MCI	F ₁ Dem
Single system								
1	43.5	72.8	57.7	0.0	42.3	73.3	53.6	0.0
2	42.3	60.3	42.4	24.2	57.3	64.9	57.1	50.0
3	43.1	64.9	35.9	28.6	40.9	72.7	33.3	16.7
4	52.9	58.9	52.4	47.4	54.5	63.4	66.7	33.3
5	38.4	60.4	37.8	17.0	47.9	57.9	42.9	42.9
6	34.7	59.0	40.0	5.0	45.9	73.7	50.0	14.3
7	39.6	63.6	44.4	10.8	47.9	73.7	53.3	16.7
Model ensemble								
Ensemble#1	64.2	76.9	62.7	52.9	67.8	77.3	69.0	57.2
Ensemble#2	63.1	75.6	58.2	55.6	67.4	78.3	66.7	57.1

as macro-descriptors and pause features are found to be very informative. In contrast, no clear patterns or trends were found in relation to the type of classifier used for their training.

Final system selection. As our final systems for the PROCESS Challenge 2025 submission, we selected the two ensembles – highlighted with a red edge in Figure 3 – that offered the best balance between UAF₁ in the training and development sets, and F1 score for the dementia class, henceforth referred to as *Ensemble #1* and *Ensemble #2*. Each ensemble consists of 6 single systems, as Table 3 describes. Notably, five systems are shared between the two ensembles. The results obtained by each of these individual systems, as well as those obtained after fusion, are presented in Table 4. The performance of the two fusion systems on the official test set is 54.36% and 59.34% for the *Ensemble#1* and *Ensemble#2*, respectively.

Limitations. Among the limitations of this work, we first highlight the challenges arising from the dataset, namely the significant imbalance between diagnostic classes and the extreme data scarcity for the dementia class (cf. Table 1). These issues may partly explain the performance gap observed between the held-out development set and the test set, as well as the lower performance compared to previous studies on the ADReSS [36, 29] or Taukadiad [9] Challenges, even considering that those targeted binary classification problems. Moreover, it is also necessary to take into account the complexity and subjectivity inherent in clinical assessments. The distribution of MMSE scores provided in the dataset exhibits considerable overlap between conditions –even between groups that are presumed to be more distinct, such as HC and Dementia–, highlighting the difficulty of differentiating between cognitive disorders in their early stages.

A second limitation is that our approach does not consider demographic data, which physicians typically consider in clinical scenarios. This omission is due to two factors: the limited size of the dataset and the unavailability of demographic infor-

mation for the test set, making its incorporation infeasible in our approach. As a result, demographic differences, such as age distributions of dementia patients between different partitions, may have had a negative impact on our models. This also has a negative impact on the “real-world” applicability of our approach and this dataset, since demographic information is crucial for accurate clinical decision making.

A third limitation is the potential influence of interviewers’ interventions on the results. An automated diagnostic tool should not be influenced by the behavior of healthcare practitioners. While this issue could be partially mitigated by diarization, our experiments with state-of-the-art diarizers failed to yield high-quality outputs. Moreover, even with diarization, the subject’s behavior remains influenced by interventions, such as prompts to recall specific information (e.g., “farm animals”). Although it is natural for individuals with cognitive impairment to require additional support during speech tasks, this dependency highlights a key limitation of the data collection method.

Finally, the potential for overfitting must be acknowledged. Despite using cross-validation on the training set and maintaining a held-out development set, the risk of overfitting increases due to the extensive comparison of approaches on a limited dataset. Future research and validation on larger datasets are necessary to address this issue.

5. Conclusions

This paper presents our contributions for the automatic assessment of cognitive decline from spontaneous speech in the context of the PROCESS Challenge [17]. We conducted extensive experimentation, addressing the challenge from multiple perspectives. This included evaluating the accuracy of various ASR models, exploring numerous acoustic- and text-based feature descriptors (and their combinations), and studying a wide range of classification approaches. Our findings suggest that fusion of complementary systems achieve higher performance. Systems based on pause-related speech features and macro-descriptors are consistently selected across model ensembles, highlighting the effectiveness of KB speech features and high-level assessment concepts as potential biomarkers of cognitive decline. Additional insights demonstrate the capability of utterance-level speech embeddings to encode and fuse information from multiple speech assessment tasks, as well as of Longformer in capturing long-term dependencies, particularly in the CTD task, where processing of context is essential. The limitations encountered, however, highlight the ongoing research challenges in developing automatic systems for cognitive decline assessment, particularly in real-world scenarios involving transitional stages, small corpora, and significant data imbalance.

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